

Md Kibria Saroare

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Summary

FE Electrical & Computer qualified Electrical Engineer with over 8 years of experience in power generation, substation maintenance, and smart grid systems. Skilled in DigSILENT PowerFactory, ETAP, Matlab/Simulink, and PSCAD for power system analysis and layout design. Currently pursuing advanced research in algorithm-system co-design for smart grid monitoring and machine learning applications.

Education

MS in Electrical Engineering, Louisiana Tech University

Expected Graduation: December 2026

Bachelor's degree in Electrical Engineering, Khulna University of Engineering & Technology

September 2014

Professional Certificates & Licenses

- Passed FE in Electrical and Computer.
- Member (MIEB-40419), The Institution of Engineers, Bangladesh.

Technical Skills

- **Tools:** DigSILENT PowerFactory, ETAP, Matlab/Simulink, PSCAD, Vitis.
- **Programming & Frameworks:** Python, C, C++, Java, Verilog, PyTorch, Optuna.
- **Code & Standards:** ANSIC2, UL, IEC, API, IEEE.
- **Core Competencies:** Power System Analysis, Grid Monitoring, Renewable Integration, Regulatory Compliance, Testing & Validation, FPGA Implementation.
- **Power System:** Design and analysis of AIS and GIS substations, substation layouts, load flow and arc flash studies, and fault analysis. Skilled in LT/HT cable sizing, termination, and conduit design. Proficient in alternator, protective relay coordination, circuit breaker selection, isolator, transformer, busbar trunking systems, control panels, power factor correction, and capacitor bank design. Experienced with SFC/SEE, BESS, UPS, VFDs, voltage regulation, commissioning of overhead and underground lines.

Awards & Recognition

- Awarded \$10,000 NSF and Louisiana Board of Regents GRASS grant for Grid monitoring proposal (Ranked 2nd/39).
- TPEC 2026 NSF Travel Grant Award.

Professional Experience

Senior Electrical Engineer, *Ashuganj Power Station Co. Ltd.* – Bangladesh

Mar. 2021 – Aug. 2025

- Improved reliability of a 1690 MW generation utility by implementing advanced diagnostics and equipment optimization.
- Strengthened regulatory and safety compliance through testing, validation, and continuous improvement initiatives.
- Enhanced grid resilience by developing cybersecurity strategies for Smart Grid and AMI integration.
- Minimized system downtime by leading cross-functional troubleshooting of high-impact electrical and control failures.

Electrical Engineer, *Ashuganj Power Station Co. Ltd.* – Bangladesh

Sept. 2016 – Mar. 2021

- Increased design accuracy and system performance through detailed power system modeling for major plant upgrades.
- Restored operational continuity by diagnosing and resolving complex electrical component failures in critical systems.
- Improved grid stability by enforcing strict safety, quality, and compliance standards across plant operations.

Lecturer, *Prime University* – Bangladesh

Oct. 2014 – Sept. 2016

- Delivered instruction in Power System Analysis, Electrical Machines, and Circuits to strengthen engineering fundamentals.
- Designed course modules and laboratory assessments that improved student competency and hands-on proficiency.

Research Experience

Research Assistant, *Louisiana Tech University* – Ruston, LA

Aug. 2025 – Present

- Advancing a Spatio-Temporal Grid Resilience Framework for Hyperscale AI Data Center Integration.
- Developing hardware-optimized algorithms for real-time grid monitoring and event detection on FPGA platforms.
- Implementing advanced machine learning techniques to enhance situational awareness in complex networks.
- Validating high-speed frameworks through rigorous simulation to ensure reliability in dynamic grid environments.

Publications

- Saroare, Md Kibria; Hasnat, Md Abul; Ahmed, Md Rubel "GridStream: A Hardware-Efficient Framework for Bandwidth-Constrained Point-on-Wave Disturbance Monitoring." TPEC 2026 (Accepted)
- Saroare, Md. Kibria; Sefat, Md. Syadus; Sen, Sajib; Shahjahan, Md. "A Modified Penalty Function in Fuzzy Clustering Algorithm." *2017 Intelligent Systems Conference (IntelliSys)*, 2017, pp. 446-451.